

A Tale of Two Phases

Order and disorder in inhomogeneous materials

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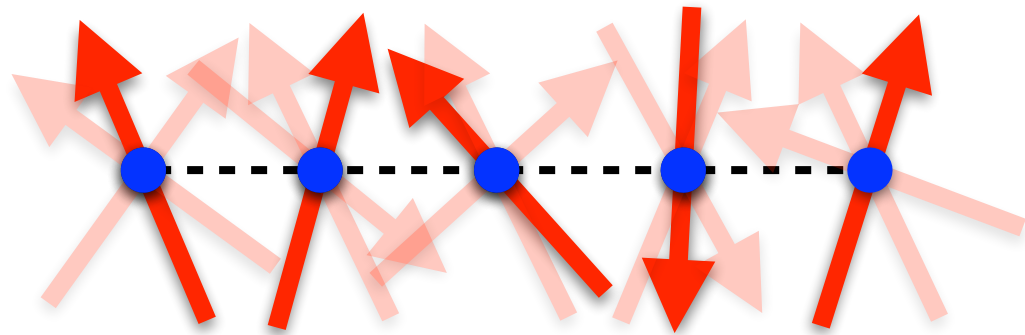
Contents

- Spontaneous Symmetry Breaking
 - Short Range vs Long Range order
 - Fundamental excitations
- Inhomogeneous material
 - Phenomenology and novel phases
 - Energetics, fluctuations and dimensionality
- An example: bond-random antiferromagnet
 - Collinear vs Non-collinear

$O(N)$ symmetric ferromagnet

$$H = J \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j$$

$$\mathbf{S} \rightarrow R \cdot \mathbf{S} \quad \forall \quad R \in O(N) \implies H \rightarrow H$$



High Temperature

$$\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle \propto e^{-\frac{|\mathbf{r}_i - \mathbf{r}_j|}{\xi(T)}}$$

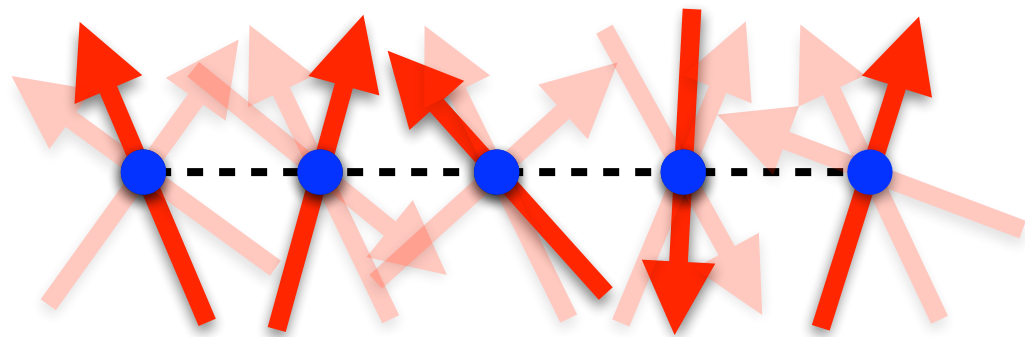
Fluctuations entropically favored at higher temperatures, T

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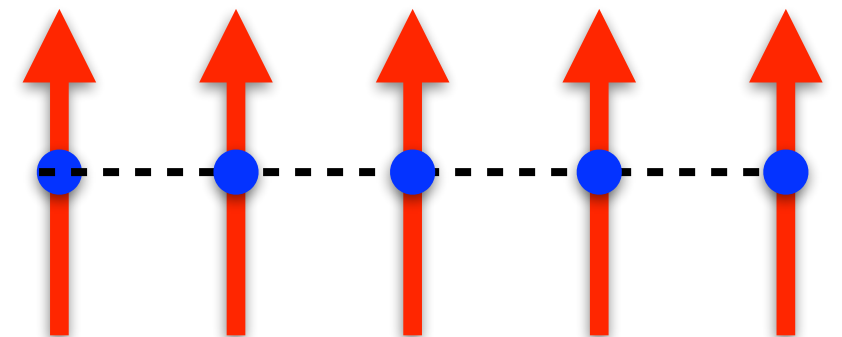
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Spontaneous Symmetry
Breaking



High Temperature

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Low Temperature

$$\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle \propto 1$$

Fluctuations entropically favored at higher temperatures, T

When?

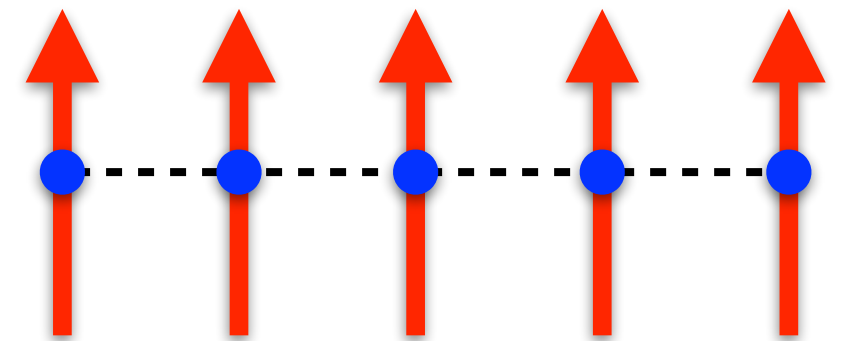
The partition function

$$Z(\mathbf{h}) = \sum_{\text{configurations}} e^{-\frac{1}{T} \left[J \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j + \mathbf{h} \cdot \mathbf{S}_i \right]}$$

is singular in the limit $\mathbf{h} \rightarrow 0$

$$\lim_{\mathbf{h} \rightarrow 0} \langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle_{\mathbf{h}} \neq \langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle_{\mathbf{h}=0}$$

Spontaneous Symmetry
Breaking



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No free energy gain from entropy at $T=0$

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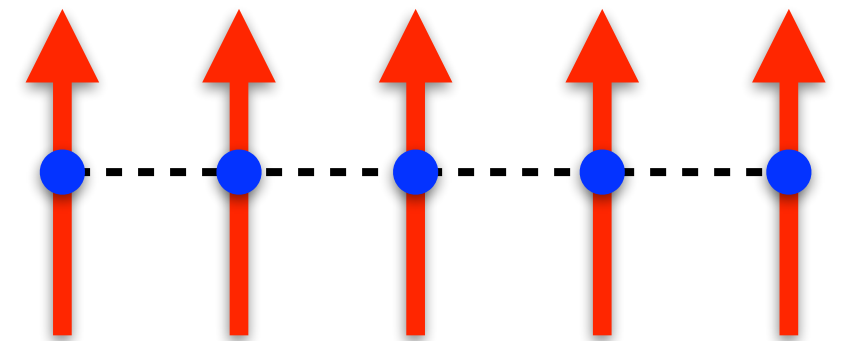
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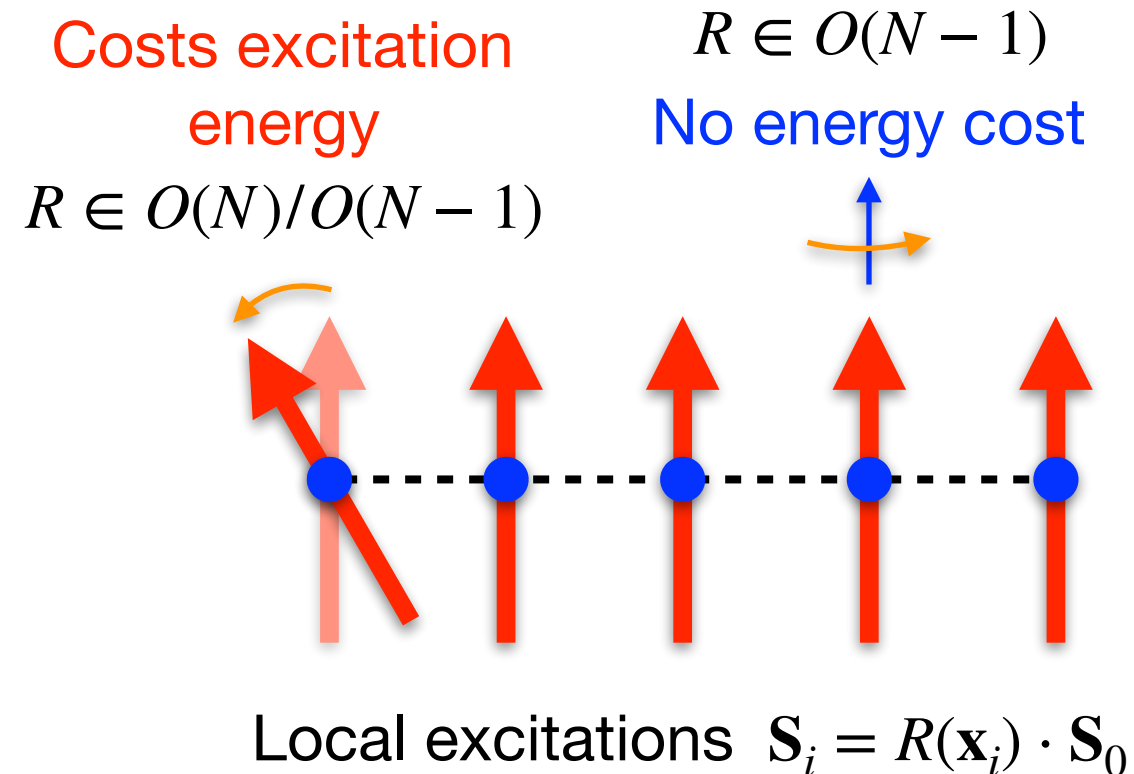
Low Temperature

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No free energy gain from entropy at $T=0$

Henceforth, we focus on $T = 0$

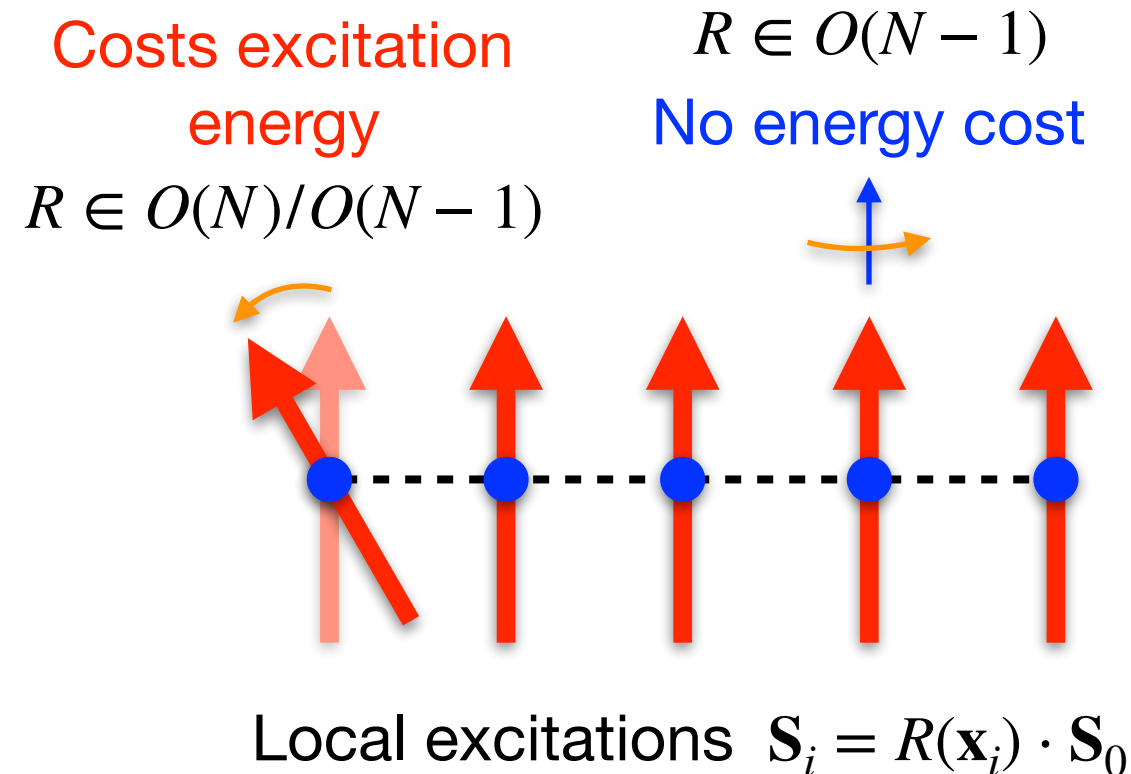
Fundamental excitations at $T = 0$



$$H = J \sum_{\langle i,j \rangle} \mathbf{S}_i \cdot \mathbf{S}_j = J \sum_{\langle i,j \rangle} \mathbf{S}_0 \cdot R^{-1}(\mathbf{x}_i) R(\mathbf{x}_j) \cdot \mathbf{S}_0$$

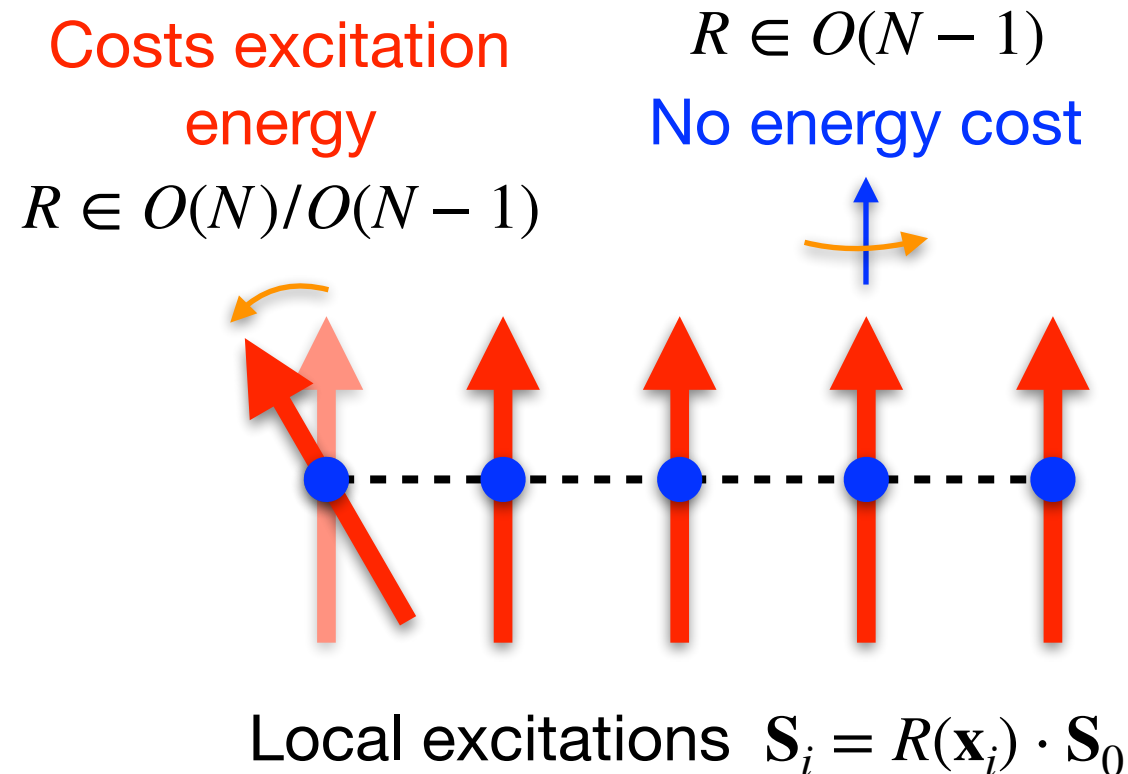
Long-wavelength
 slow deformation of
 $R(\mathbf{x}) \in O(N)/O(N-1)$

Fundamental excitations at $T = 0$



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Ripple like excitations with a gapless spectrum $E \propto \rho k^2$

Goldstone Modes!

Excitations of symmetry breaking G/H

G - invariant model with H - invariant ordered phase : $\langle \phi \rangle = \phi_0$

$$gh\phi_0 = g\phi_0 \quad \forall \quad g \in G \text{ and } h \in H$$

The Lagrangian :

$$\mathcal{L} = \rho \left[\partial_\mu \phi \cdot \partial^\mu \phi + \lambda \left(|\phi|^2 - \phi_0^2 \right) \right]$$

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Excitations around to the ordered state :

Goldstone
Mode

← Phase Modulation

$$\phi = R(x) \cdot (\phi_0 + \mathbf{H}(x))$$

Amplitude Modulation →

Higgs
Mode

Excitations of symmetry breaking G/H

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The Lagrangian :

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Excitations around to the ordered state :

The Higgs mode is massive and unstable!

Massless Goldstone mode Lagrangian for elementary excitations

$$\mathcal{L} = \rho \text{Tr} \left[\partial_\mu R^{-1} \partial^\mu R \right] + \text{higher derivative terms}$$

With $R \in G/H$

Inhomogeneous materials

A generic inhomogeneous ferromagnet

$$H = \sum_{\langle i,j \rangle} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j + \mathbf{h}_i \cdot \mathbf{S}_i$$

Random couplings

Random fields

The diagram illustrates the concept of inhomogeneous materials. At the top, the title 'Inhomogeneous materials' is displayed in blue. Below it, the text 'A generic inhomogeneous ferromagnet' is shown. The central part of the diagram features the Hamiltonian equation $H = \sum_{\langle i,j \rangle} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j + \mathbf{h}_i \cdot \mathbf{S}_i$. A blue arrow points from the text 'Random couplings' to the coupling term J_{ij} in the equation. A red arrow points from the text 'Random fields' to the field term $\mathbf{h}_i$. At the bottom, a large blue arrow points from the text 'Random system parameters' to the text 'More fluctuations over ordered state'.

Random system parameters



More fluctuations over
ordered state

Inhomogeneous materials

A generic inhomogeneous ferromagnet

$$H = \sum_{\langle i,j \rangle} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j + \mathbf{h}_i \cdot \mathbf{S}_i$$

Random couplings

Random fields

Random system parameters



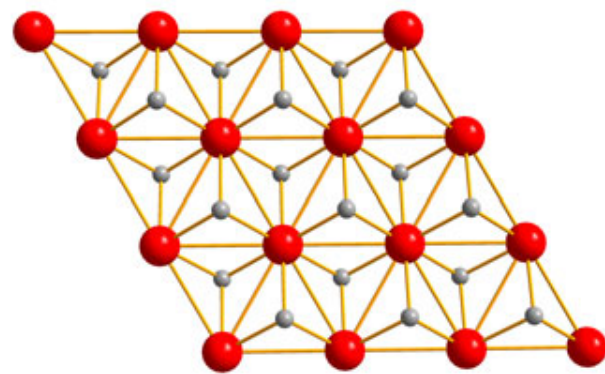
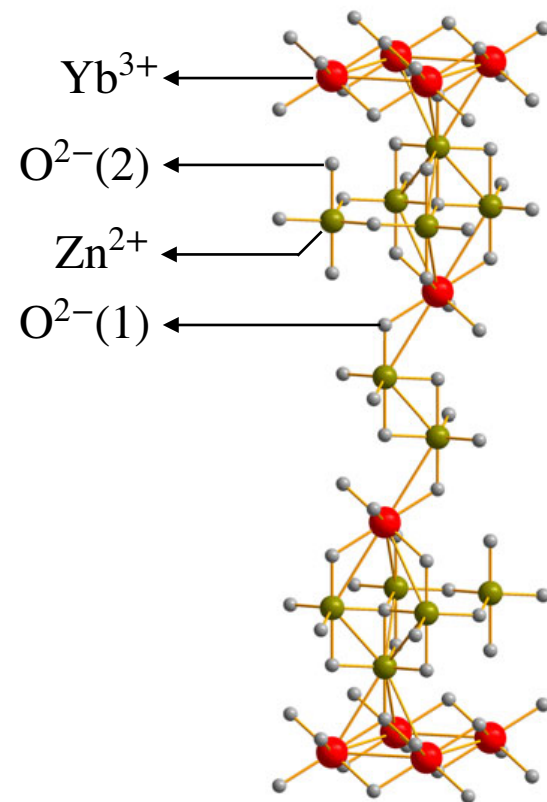
More fluctuations over
ordered state

Possible outcomes :

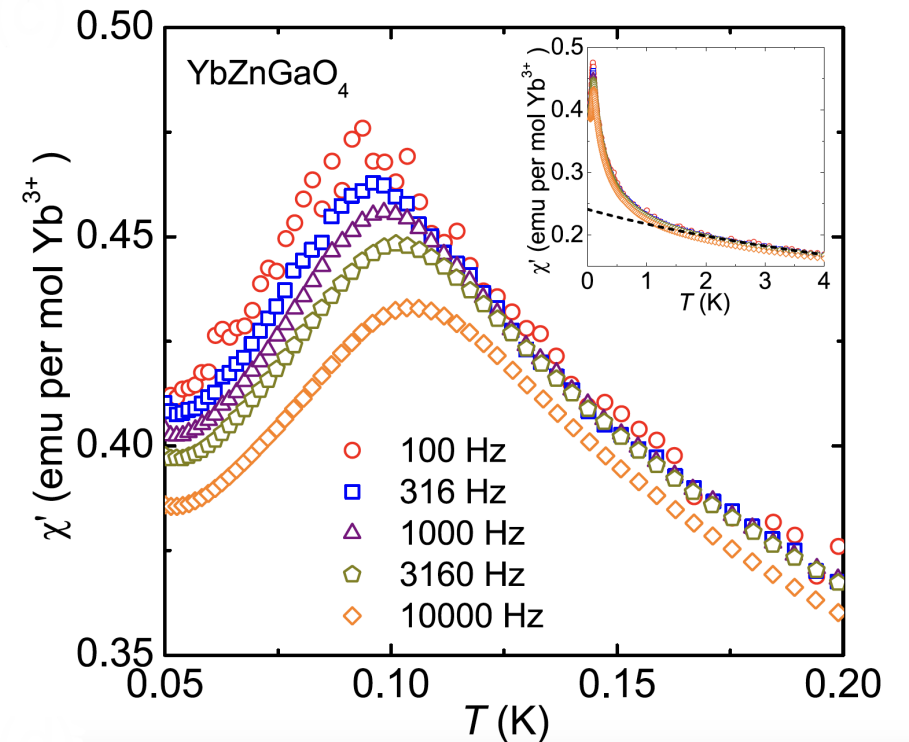
- Disorder driven quantum phase transitions
- Disordered glass phases (multiple timescales)
- Disordered phases with quasi-stable excitations (like quantum liquids)

Inhomogeneous materials

Triangular lattice antiferromagnet YbZnGaO_4



No magnetic order up to very low temperatures

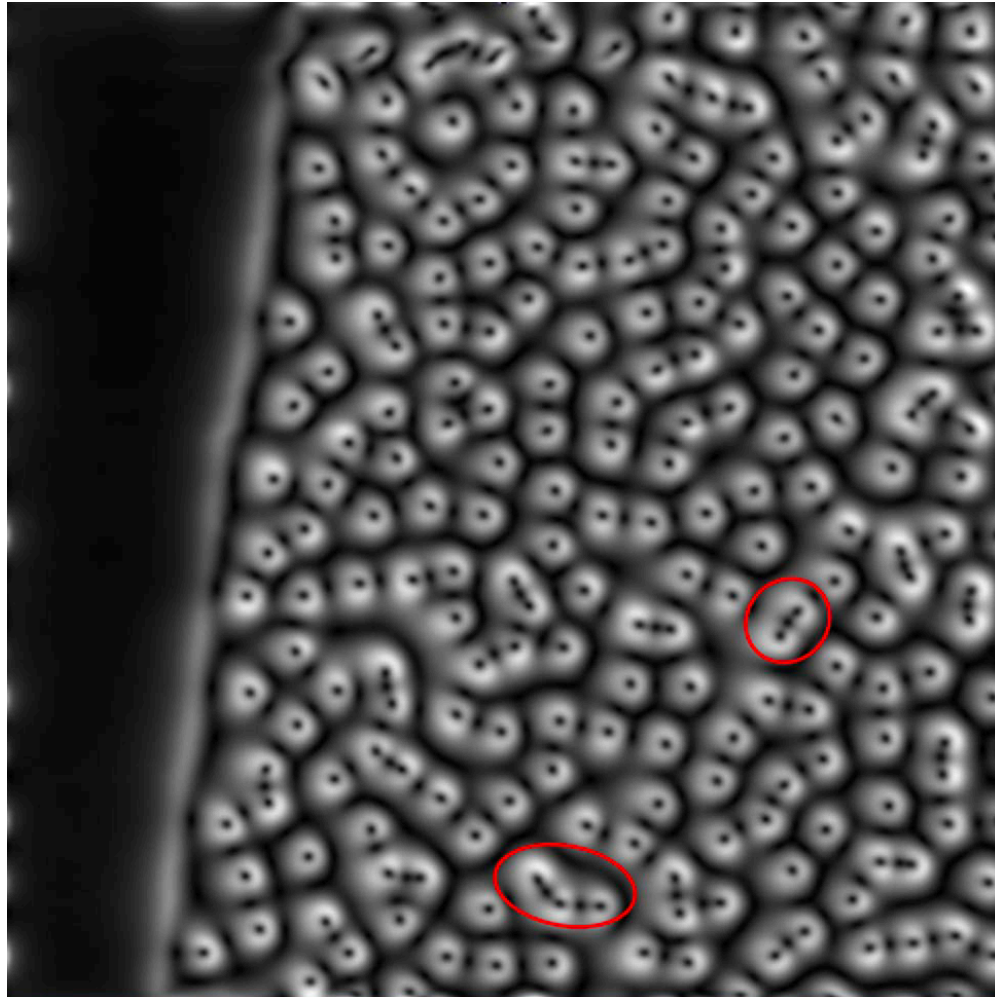


Low frequency peaks of spin susceptibility hints glassy dynamics

Spin-glass phase

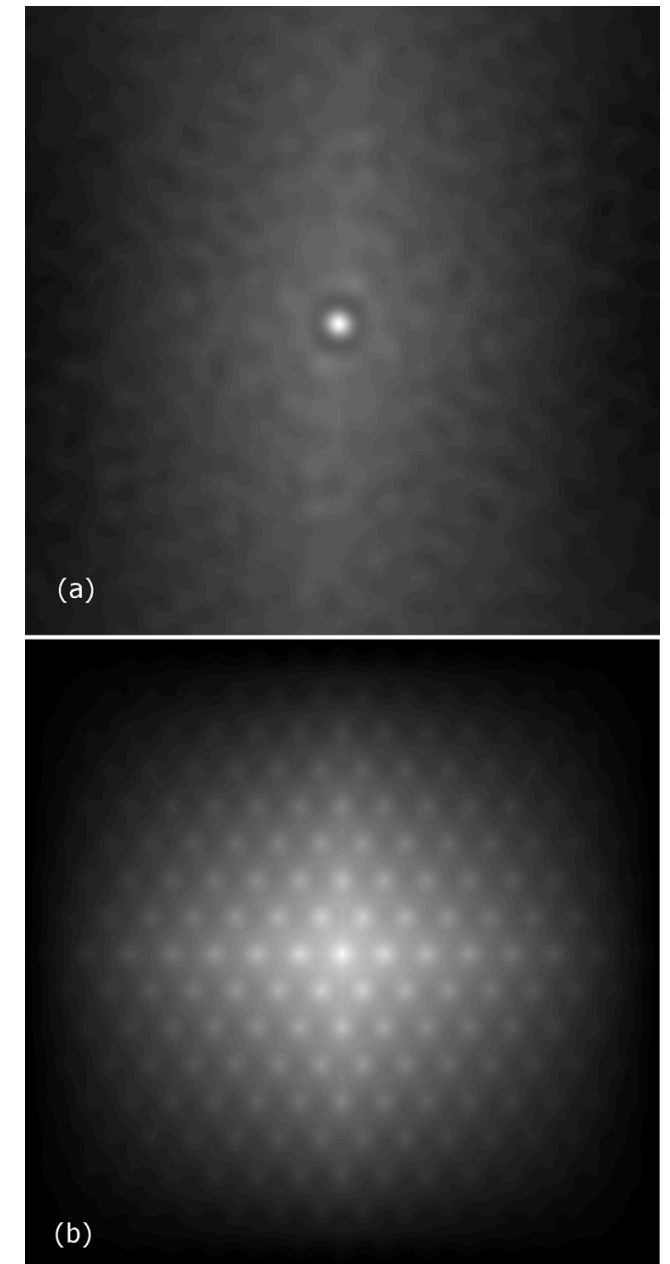
Inhomogeneous materials

Dirty type-II superconductor $\text{YBa}_2\text{Cu}_3\text{O}_{7-x}$ (YBCO)



Super-currents around randomly distributed Vortex cores

Vortex-glass phase



Auto-correlation showing lack of long range order in comparison to an ideal triangular lattice

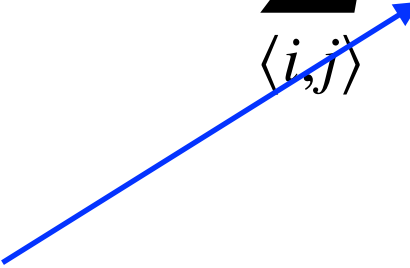
Energetics of inhomogeneous media

Goldstone modes couple directly to the random parameters

$$\mathbf{S}_i = R(\mathbf{x}_i) \cdot \mathbf{S}_0 \text{ with } R \in G/H$$

$$H = \sum_{\langle i,j \rangle} J_{ij} \mathbf{S}_i \cdot \mathbf{S}_j + \mathbf{h}_i \cdot \mathbf{S}_i$$

Random couplings



- Breaks translational symmetry
- Does not break any other symmetry

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Random couplings

- Breaks translational symmetry
- Does not break any other symmetry

Random fields

- Breaks translational symmetry
- Also breaks time-reversal, spin inversion and other symmetries

Energetics of inhomogeneous media

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An inhomogeneous spatio-temporal, $x = (\tau, \mathbf{x})$ Lagrangian :

$$\mathcal{L}(x) = \rho \text{Tr} \left[\partial_\mu R^{-1}(x) \partial^\mu R(x) \right] \quad \text{Bulk Goldstone mode}$$

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$$+ \rho \delta J_\alpha(\mathbf{x}) \text{Tr} \left[Q_\mu^\alpha \left(R^{-1}(x) \partial^\mu R(x) \right) \right] \quad \text{Symmetry-obeying bond-randomness coupling}$$

Energetics of inhomogeneous media

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$$\begin{aligned} \mathcal{L}(x) = & \rho \text{Tr} \left[\partial_\mu R^{-1}(x) \partial^\mu R(x) \right] && \text{Bulk Goldstone mode} \\ & + \rho \delta J_\alpha(\mathbf{x}) \text{Tr} \left[Q_\mu^\alpha \left(R^{-1}(x) \partial^\mu R(x) \right) \right] && \text{Symmetry-obeying bond-} \\ & && \text{randomness coupling} \\ & + \rho \delta h_\alpha(\mathbf{x}) \text{Tr} \left[P^\alpha R(x) \right] && \text{Symmetry-breaking field-} \\ & && \text{randomness coupling} \end{aligned}$$

Where Q_μ^α and P^α are projection matrices coupling the random parameters to select Goldstone modes.

Lower critical dimension: Random field

$$\mathcal{L}(x) = \rho \text{Tr} \left[\partial_\mu R^{-1}(x) \partial^\mu R(x) \right] + \rho \delta h_\alpha(\mathbf{x}) \text{Tr} \left[P^\alpha R(x) \right]$$

Goldstone mode correlation in tree-level perturbation theory :

$$C(|\mathbf{x} - \mathbf{y}|) = \langle \text{Tr} [R^{-1}(x) R(y)] \rangle$$

$$\propto \frac{1}{\rho} \int \frac{d^d k}{(2\pi)^d} \int \frac{d\omega}{2\pi} \frac{1}{\omega^2 + \mathbf{k}^2} e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} + \int \frac{d^d k}{(2\pi)^d} \int \frac{d^d k'}{(2\pi)^d} \frac{h_\alpha(\mathbf{k}) h_\beta(\mathbf{k}')}{\mathbf{k}^2 \mathbf{k}'^2} e^{i(\mathbf{k} \cdot \mathbf{x} + \mathbf{k}' \cdot \mathbf{y})}$$

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For uncorrelated random fields :

$$\overline{h_\alpha(\mathbf{x}) h_\beta(\mathbf{x}')} = \Delta^2 \delta_{\alpha\beta} \delta^{(d)}(\mathbf{x} - \mathbf{x}')$$

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Imry and Ma, Phys. Rev. Lett. **35**, 1399 (1975) Infrared divergent for $d \leq 4$

Lower critical dimension: Random field

$$\mathcal{L}(x) = \rho \text{Tr} \left[\partial_\mu R^{-1}(x) \partial^\mu R(x) \right] + \rho \delta h_\alpha(\mathbf{x}) \text{Tr} \left[P^\alpha R(x) \right]$$

- Goldstone mode fluctuation is divergent for $d \leq 4$.
- This prohibits spontaneous symmetry breaking.

Lower critical dimension for random field systems
with continuous symmetry is 4

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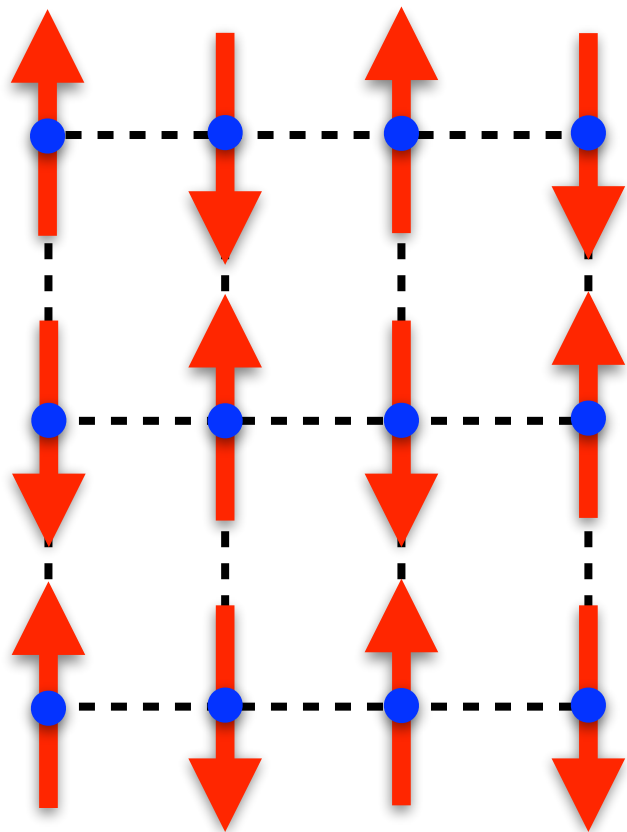
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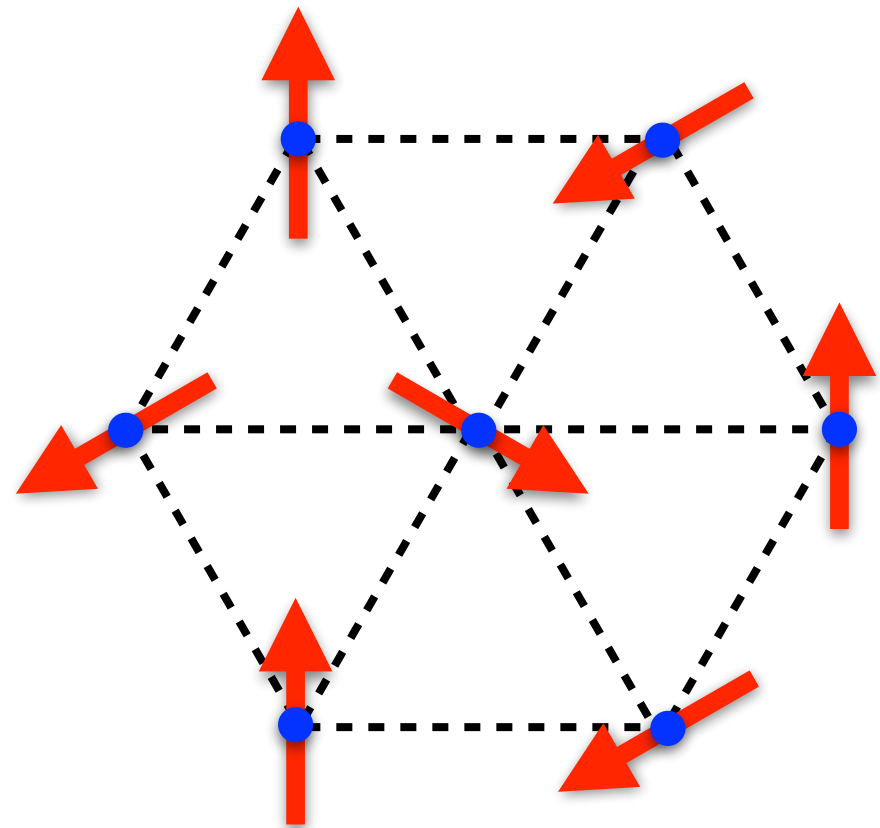
- The resulting short-ranged-ordered state may or may not have glassy dynamics.
- Full characterization of the disordered state requires beyond tree level analysis.

Lower critical dimension: Random bond

Collinear magnets
e.g. square-lattice antiferromagnet
SLAF

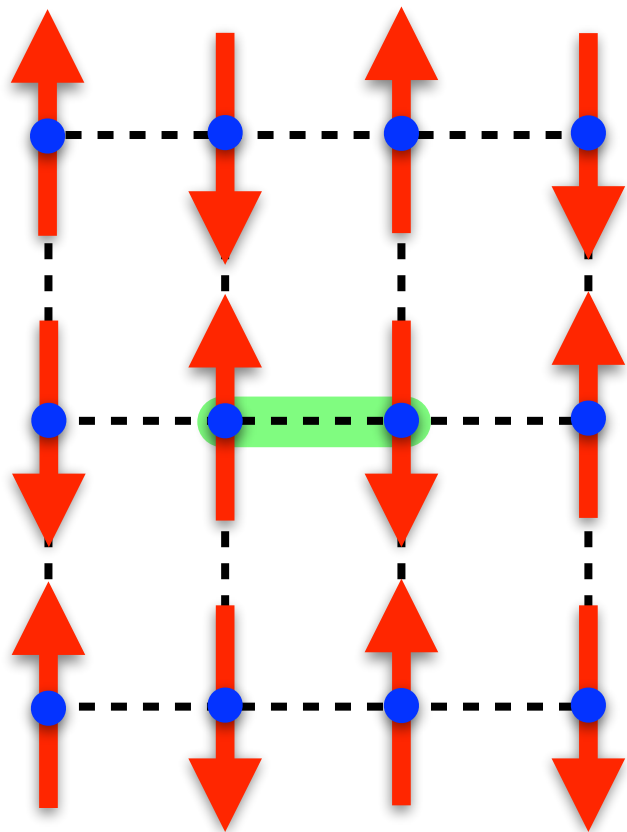


Non-collinear magnets
e.g. triangular-lattice antiferromagnet
TLAF



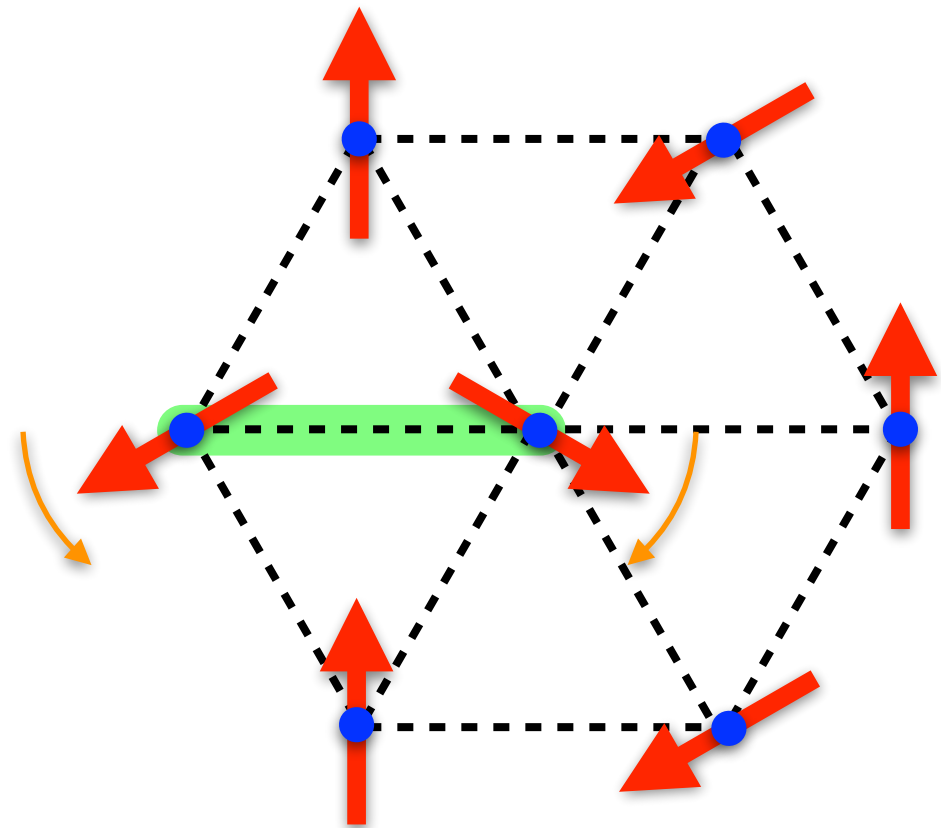
Lower critical dimension: Random bond

Collinear magnets
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SLAF



Spins do not respond to bond impurity

Non-collinear magnets
e.g. triangular-lattice antiferromagnet
TLAF



Spins reorient as frustration is released locally with bond impurity

Lower critical dimension: Random bond

Collinear magnets

e.g. square-lattice antiferromagnet
SLAF

Randomness couples to higher order in
Goldstone mode expansion

$$\mathcal{L}(x) = \rho \text{Tr} \left[\partial_\mu R^{-1}(x) \partial^\mu R(x) \right] \\ - \rho \delta J(\mathbf{x}) \text{Tr} \left[\partial_\mu R^{-1}(x) \partial^\mu R(x) \right]$$

Non-collinear magnets

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- The replica trick : Spatially random random action to homogeneous action with multiple copies.
- Allows us to carry on renormalization group analysis on these models.

Lower critical dimension: Random bond

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SLAF

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This coupling is irrelevant in the RG sense.

Collinear ordering is
robust against small bond
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e.g. triangular-lattice antiferromagnet
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The coupling is marginal in $d = 2$

Non-collinear ordering is
fragile against small bond
randomness in $d \leq 2$

Random bond non-collinear magnets

$$\mathcal{L}(x) = \rho \text{Tr} \left[\partial_\mu R^{-1}(x) \partial^\mu R(x) \right] + \rho \delta J_\alpha(\mathbf{x}) \text{Tr} \left[Q_\mu^\alpha \left(R^{-1}(x) \partial^\mu R(x) \right) \right]$$

Goldstone mode correlation in tree-level perturbation theory :

$$C(|\mathbf{x} - \mathbf{y}|) = \langle \text{Tr} [R^{-1}(x) R(y)] \rangle \\ \propto \frac{1}{\rho} \int \frac{d^d k}{(2\pi)^d} \int \frac{d\omega}{2\pi} \frac{1}{\omega^2 + \mathbf{k}^2} e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} + \int \frac{d^d k}{(2\pi)^d} \int \frac{d^d k'}{(2\pi)^d} \frac{h_\alpha(\mathbf{k}) h_\beta(\mathbf{k}')}{|\mathbf{k}| |\mathbf{k}'|} e^{i(\mathbf{k} \cdot \mathbf{x} + \mathbf{k}' \cdot \mathbf{y})}$$

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Goldstone mode correlation in tree-level perturbation theory :

$$C(|\mathbf{x} - \mathbf{y}|) = \langle \text{Tr} [R^{-1}(x) R(y)] \rangle \\ \propto \frac{1}{\rho} \int \frac{d^d k}{(2\pi)^d} \int \frac{d\omega}{2\pi} \frac{1}{\omega^2 + \mathbf{k}^2} e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} + \int \frac{d^d k}{(2\pi)^d} \int \frac{d^d k'}{(2\pi)^d} \frac{h_\alpha(\mathbf{k}) h_\beta(\mathbf{k}')}{|\mathbf{k}| |\mathbf{k}'|} e^{i(\mathbf{k} \cdot \mathbf{x} + \mathbf{k}' \cdot \mathbf{y})}$$

For un-correlated random bonds :

$$C(|\mathbf{x} - \mathbf{y}|) \propto \frac{1}{\rho} \int \frac{d^d k}{(2\pi)^d} \int \frac{d\omega}{2\pi} \frac{1}{\omega^2 + \mathbf{k}^2} e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} + \Delta^2 \int \frac{d^d k}{(2\pi)^d} \frac{1}{\mathbf{k}^2} e^{i\mathbf{k} \cdot (\mathbf{x} - \mathbf{y})} \\ \propto \text{const} + \Delta^2 \log \left(\frac{|\mathbf{x} - \mathbf{y}|}{\xi_\Delta} \right) + \text{contributions from higher order corrections}$$

Random bond non-collinear magnets

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Short range correlation length ξ_Δ

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Renormalization group flow of random coupling :

$$\beta(\Delta^2) = (d - 2)\Delta^2 + C_{G/H}\Delta^4$$

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For $G = O(N)$ we have $C_{G/H} \propto (N - 2)$

So for $O(2)$ planar Heisenberg model the coupling is exactly marginal

Ground state of planar $O(2)$ model

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Ground state of planar O(2) model

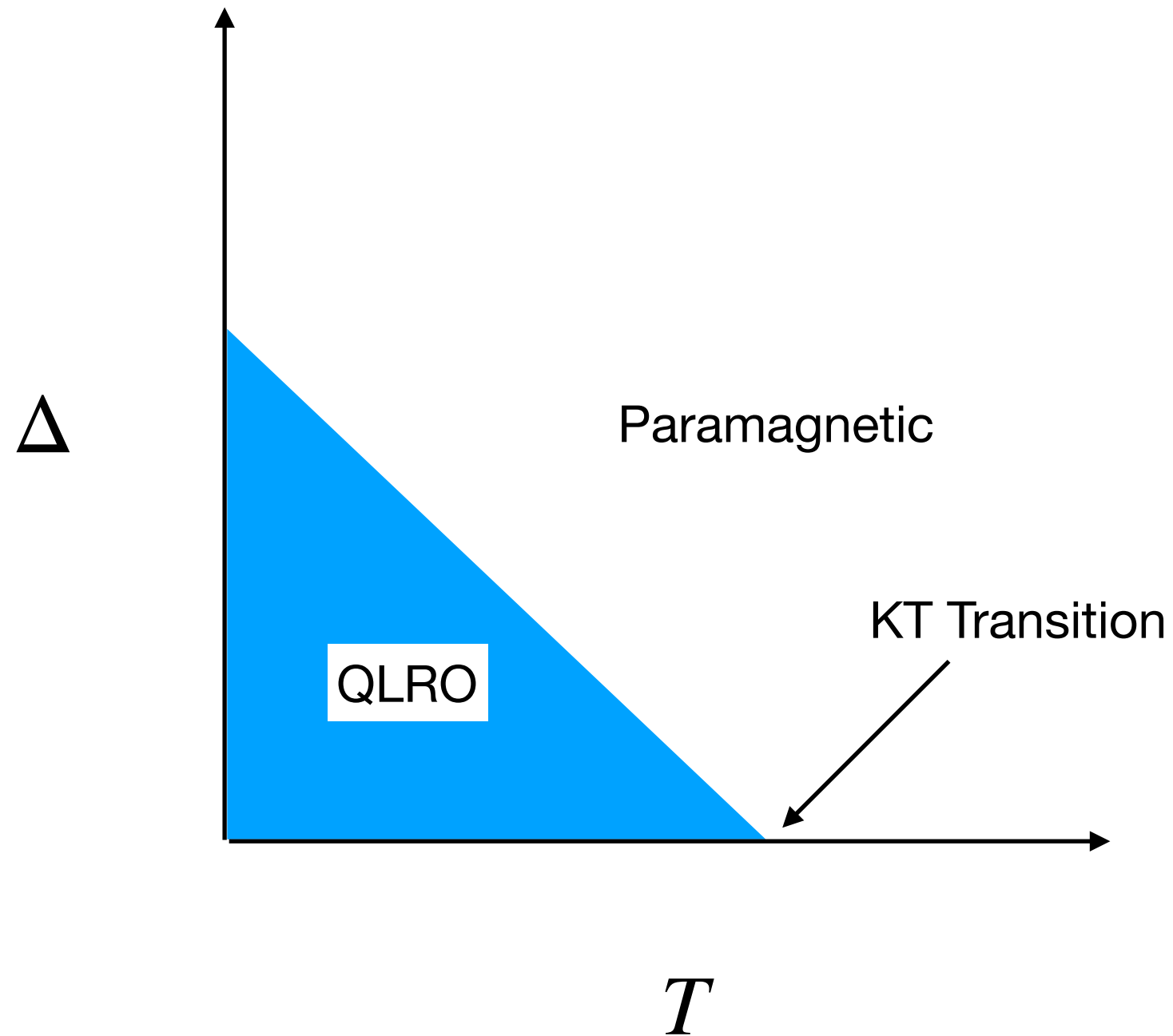
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A quasi long range order with power law
decay of correlation function

Phase diagram of planar O(2) model



Random bond: Outlook

- Character of disordered ground state for $O(N)$, $N > 2$
- Coupling of random bond with topological defects
- **KT transition driven by randomness alone?**
- Extension superconducting models

Thank You